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FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
MATHEMATICS EXTENSION
2019
YEAR 11 Yearly Examination
ANSWERS COVER SHEET**

Student Number: _____

QUESTION	MARK	ME11-1	ME11-2	ME11-3	ME11-4	ME11-5	ME11-6	ME11-7
1-5	/5	✓	✓					✓
6	/20	✓	✓					✓
7	/13	✓	✓					✓
8	/25		✓					✓
9	/28					✓		✓
10	/14	✓	✓			✓		✓
TOTAL								
	/105	/52	/77			/42		/105

Year 11 Mathematics Extension 1 outcomes

A student:

- ME11-1** uses algebraic and graphical concepts in the modelling and solving of problems involving functions and their inverses
- ME11-2** manipulates algebraic expressions and graphical functions to solve problems
- ME11-3** applies concepts and techniques of inverse trigonometric functions and simplifying expressions involving compound angles in the solution of problems
- ME11-4** applies understanding of the concept of a derivative in the solution of problems, including rates of change, exponential growth and decay and related rates of change
- ME11-5** uses concepts of permutations and combinations to solve problems involving counting or ordering
- ME11-6** uses appropriate technology to investigate, organise and interpret information to solve problems in a range of contexts
- ME11-7** communicates making comprehensive use of mathematical language, notation, diagrams and graphs



GIRRAWEE HIGH SCHOOL

MATHEMATICS EXTENSION

YEAR 11

YEARLY EXAMINATION

• **Instructions**

- Reading Time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For Section I, use the multiple-choice answer sheet
- For questions in section II, show relevant mathematical reasoning and /or calculations
- Marks may be deducted for careless or badly arranged work

Section I (5 marks)

Answer Questions 1 – 5

Use the multiple-choice answer sheet on page 5 for questions 1 – 5

1. Which of the following is the inverse function of $f(x) = 2x^{\frac{1}{5}}$?

A. $f^{-1}(x) = \frac{x^{\frac{1}{5}}}{32}$

B. $f^{-1}(x) = \frac{x^5}{32}$

C. $f^{-1}(x) = \frac{x^5}{16}$

D. $f^{-1}(x) = \frac{x^5}{2}$

2. The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x - 2$.
What is the value of k ?

A. 2

B. 12

C. 20

D. 36

3. How many arrangements of the word *OLYMPIC* are possible if the *C* and the *L* are to be together in any order?

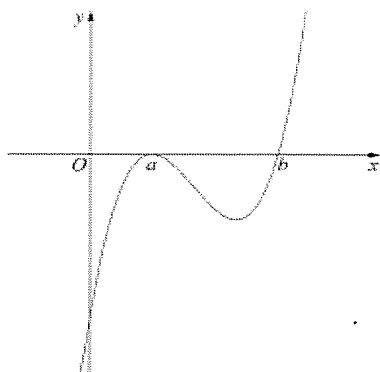
A. $5!$

B. $6!$

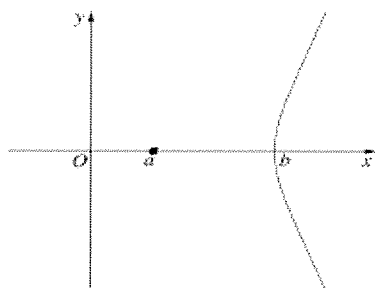
C. $2 \times 5!$

D. $2 \times 6!$

4. The diagram shows the graph of the function $y = f(x)$.



A second graph is obtained from the graph of $y = f(x)$.



Which equation best represents the second graph?

- A. $y^2 = |f(x)|$
 - B. $y^2 = f(x)$
 - C. $y = \sqrt{f(x)}$
 - D. $y = f(\sqrt{x})$
5. A curve is represented by the following parametric equations $x = t + \frac{1}{t}$, $y = t^2 + \frac{1}{t^2}$.

The Cartesian equation of the curve is:

- A. $y^2 = x - 2$
- B. $y^2 - x^2 = 2$
- C. $y = x^2 - 2$
- D. $y = x - 2$

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☐	D	☐
2.	A	☐	B	☐	C	☐	D	☐
3.	A	☐	B	☐	C	☐	D	☐
4.	A	☐	B	☐	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☐

Section II (102 marks)

- Answer questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 6 (20 marks)

a. Solve

(i) $|3x - 5| \leq 4$ [2]

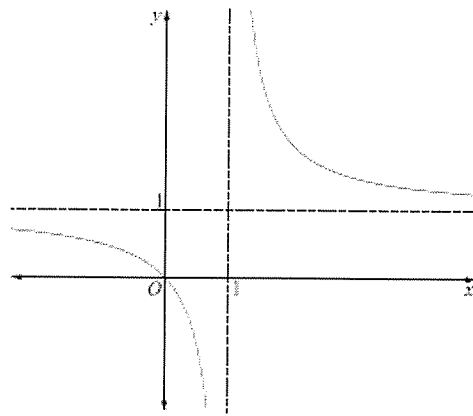
(ii) $|5x + 4| \geq 6$

[3]

$$\text{(iii)} \quad \frac{x^2 + 5}{x} > 6 \quad [3]$$

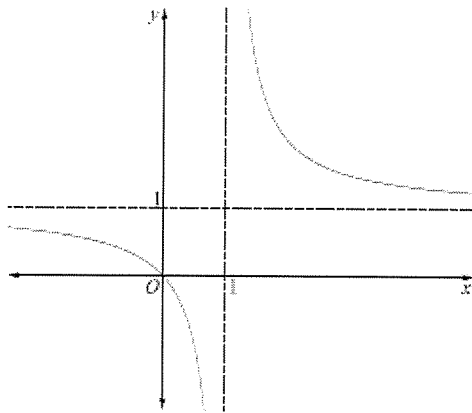
This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper appears to be a standard notebook page or a sheet of stationery.

- c. The diagram shows the graph of the function $f(x) = \frac{x}{x-1}$.

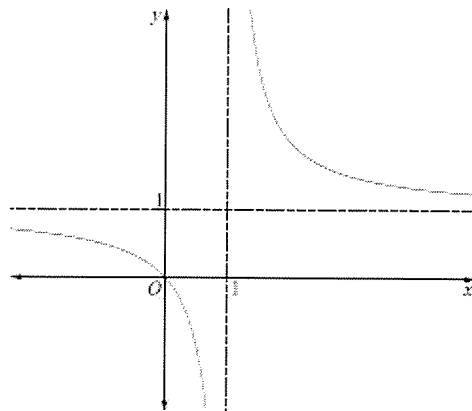


Use the graph of $y = f(x)$ to draw graphs the following transformations on the number planes provided. Clearly show all intercepts and asymptotes.

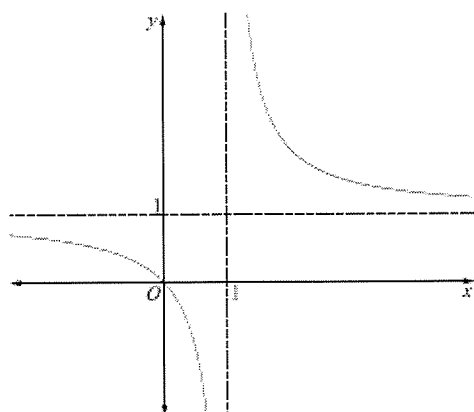
- (i) $y = |f(x)|$ [3]



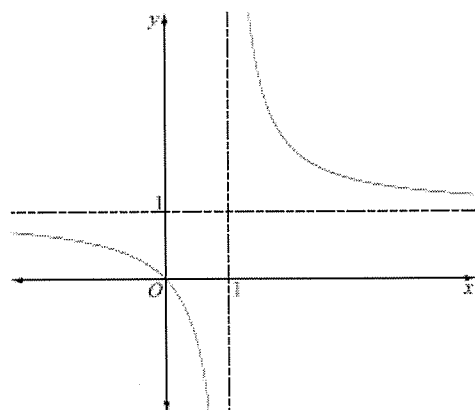
- (ii) $y = (f(x))^2$ [3]



(iii) $y = \sqrt{f(x)}$ [3]

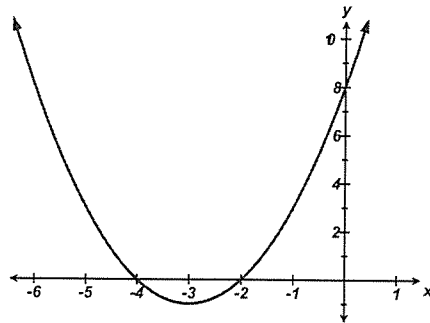


(iv) $y = f(|x|)$ [3]



Question 7 (13 marks)

- a. The graph of $f(x) = x^2 + 6x + 8$ is shown below.



- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function.

[1]

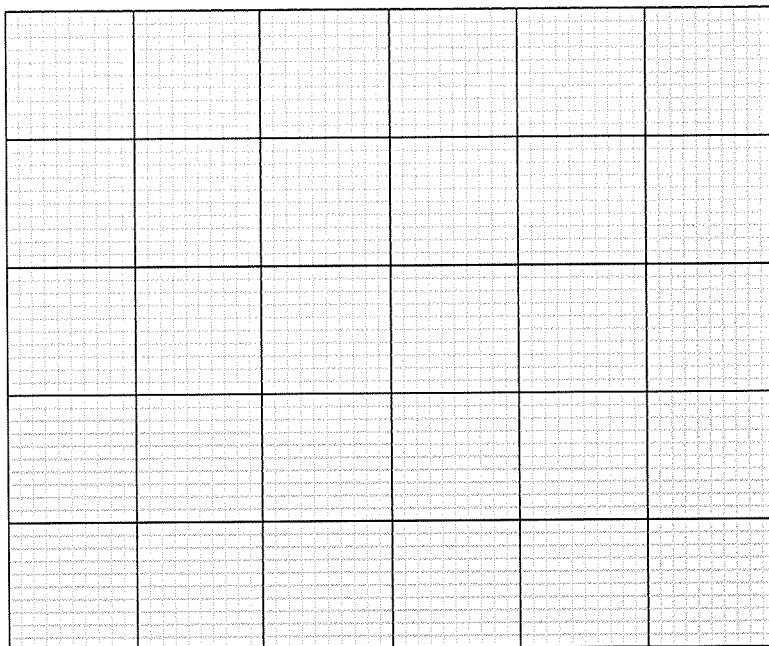
- (ii) Consider $f(x) = x^2 + 6x + 8$ for $x \geq -3$. Find the equation of $f^{-1}(x)$.

[3]

- (iii) State the domain of $f^{-1}(x)$

[1]

- (iv) Sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$ for $x \geq -3$ on the grid below, clearly showing the important features. [3]



- b. The parametric equation of a graph are:

$$x = 1 - \frac{1}{t}$$

$$y = 1 + \frac{1}{t}$$

- (i) Find the Cartesian equation of the graph. [3]

(ii) Sketch the graph.

[2]



Question 8 (25 marks)

a. If α, β, γ are the roots of $2x^3 - 3x^2 + 6x + 8 = 0$, find the value of:

(i) $\alpha + \beta + \gamma$ [1]

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ [1]

(iii) $\alpha\beta\gamma$ [1]

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]

(ii) Find the other zeros.

[3]

d. Factorise $2x^3 - 5x^2 - 4x + 12 = 0$ if it has a zero of multiplicity 2 at $x = 2$. [4]

e. Let $P(x) = (x + 1)(x - 3)Q(x) + ax + b$ where $Q(x)$ is a polynomial and a and b are real numbers. The polynomial $P(x)$ has a factor of $x - 3$.
When $P(x)$ is divided by $x + 1$, the remainder is 8.

(i) Find the values of a and b . [3]

(ii) Find the remainder $P(x)$ is divided by $(x + 1)(x - 3)$. [1]

Question 9 (28 marks)

a. Simplify

(i) $n! + (n - 1)!$ [2]

(ii) $\frac{n!}{(n - 1)!} \cdot \frac{(n - 2)!}{(n + 1)!}$ [3]

b. Siya has five different coloured blocks. She stacks two, three, four or five blocks on top of one another to form a vertical tower.

(i) How many different towers can she form that are three blocks high? [1]

(ii) How many different towers can she form in total? [2]

c. (i) In how many different ways can the letters of the word ESTEEM be arranged? [1]

(ii) In how many ways can the letters of the word ESTEEM be arranged so that the E's are together? [2]

d. Find the number of ways in which 3 boys and 4 girls line up in a queue so that girls occupy the middle position and exactly one of the two end positions. [3]

e. (i) In how many ways can a committee of 7 be selected from a group of 18 people? [1]

(ii) In how many ways can a committee of 3 men and 4 women be selected if there are 8 men and 10 women in this group? [2]

(iii) In how many ways can a committee of 3 men and 4 women be selected from this group if particular man and a particular woman can't be in the committee together? [3]

f. Six men and six women are to be seated at a round table.
In how many ways can they be seated:

(i) if there are no restrictions? [1]

(ii) if men and women alternate? [2]

(iii) if two particular people are not to sit together?

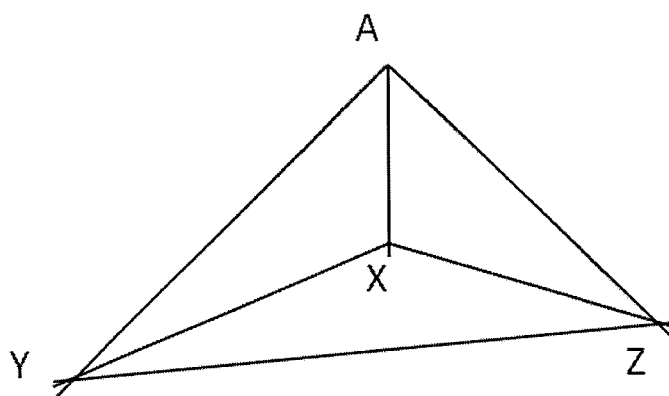
[3]

(iv) if three particular people are to sit together?

[2]

Question 10 (15 marks)

a.



The tower AX, of height h metres, is built on flat ground. X, Y and Z are points on the same horizontal plane.

From Y and Z, the angles of elevation to the top of the tower are 28° and 40° respectively. Give that $YZ = 100\text{m}$ and $\angle YXZ = 90^\circ$,

(i) show that $XY = h \cot 28^\circ$ and find a similar statement for XZ.

[2]

(ii) find the height of the tower to the nearest metre.

[3]

b. Find the inverse of $y = 3\log_e(2x + 5) - 4$.

[2]

c. Solve $x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 = 0$ if it has a root of multiplicity 4.

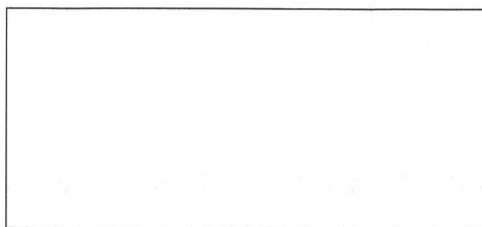
[4]

- d. A hostel has four vacant rooms. Each room can accommodate a maximum of four people. In how many ways can six people be accommodated in the four rooms? [3]

End of Examination

[illegible]

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☒ B. $f^{-1}(x) = \frac{x^5}{32}$

C. $f^{-1}(x) = \frac{x^5}{16}$

D. $f^{-1}(x) = \frac{x^5}{2}$

$$\begin{aligned} y &= 2x^{\frac{1}{5}} \\ f^{-1}(x) &; x = 2y^{\frac{1}{5}} \\ y^{\frac{1}{5}} &= \frac{x}{2} \\ y &= \frac{x^5}{32} \end{aligned}$$

2. The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x - 2$.

What is the value of k ?

A. 2

B. 12

☒ C. 20

D. 36

$$\begin{aligned} P(2) &= 2^3 - 4(2)^2 - 6(2) + k = 0 \\ k &= 20 \end{aligned}$$

3. How many arrangements of the word *OLYMPIC* are possible if the *C* and the *L* are to be together in any order?

A. $5!$

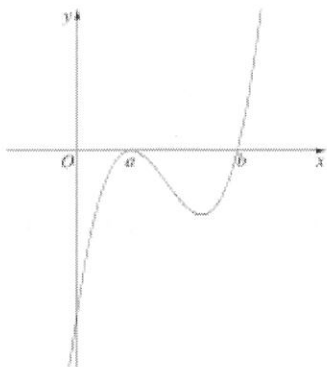
B. $6!$

C. $2 \times 5!$

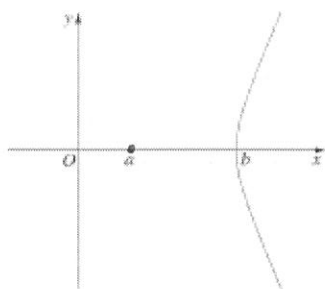
☒ D. $2 \times 6!$

$$\boxed{CL} O Y M P I$$

4. The diagram shows the graph of the function $y = f(x)$.



A second graph is obtained from the graph of $y = f(x)$.



D: $x = a, x \geq b \Rightarrow$ not A, D
 $y = \pm \sqrt{f(x)}$
 $\therefore y^2 = f(x)$

Which equation best represents the second graph?

- A. $y^2 = |f(x)|$
 B. $y^2 = f(x)$
 C. $y = \sqrt{f(x)}$
 D. $y = f(\sqrt{x})$
5. A curve is represented by the following parametric equations $x = t + \frac{1}{t}$, $y = t^2 + \frac{1}{t^2}$.
 The Cartesian Equation of the curve is:

- A. $y^2 = x - 2$
 B. $y^2 - x^2 = 2$
 C. $y = x^2 - 2$
 D. $y = x - 2$

$x^2 = \left(t + \frac{1}{t}\right)^2$
 $= t^2 + 2 + \frac{1}{t^2}$
 $\therefore x^2 - 2 = t^2 + \frac{1}{t^2}$
 $= y$

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☒	C	☐	D	☐
2.	A	☐	B	☐	C	☒	D	☐
3.	A	☐	B	☐	C	☐	D	☒
4.	A	☐	B	☒	C	☐	D	☐
5.	A	☐	B	☐	C	☒	D	☐

Section II (100 marks)

- Answer questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
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Question 6 (20 marks)

a. Solve

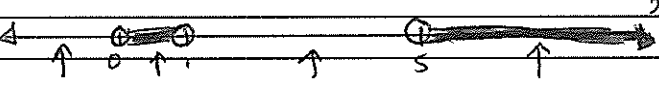
(i) $|3x - 5| \leq 4$ [2]

$$\begin{aligned} -4 &\leq 3x - 5 \leq 4 \\ 1 &\leq 3x \leq 9 \\ \frac{1}{3} &\leq x \leq 3 \end{aligned}$$

(ii) $|5x + 4| \geq 6$ [3]

$$\begin{aligned} 5x + 4 &\leq -6 & \text{or} & & 5x + 4 &\geq 6 \\ 5x &\leq -10 & & & 5x &\geq 2 \\ x &\leq -2 & & & x &\geq \frac{2}{5} \end{aligned}$$

(iii) $\frac{x^2 + 5}{x} > 6$ [3]

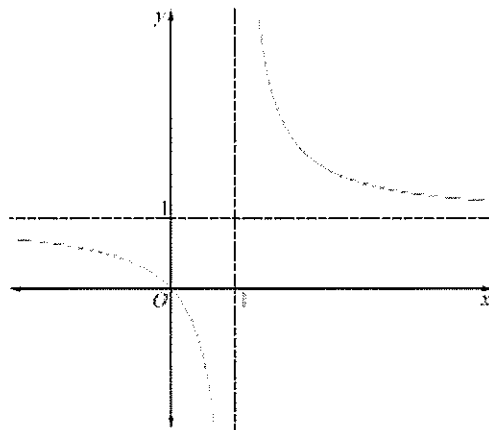
$$\begin{aligned} x \neq 0 \quad \text{so solve} \quad \frac{x^2 + 5}{x} &= 6 \\ x^2 + 5 &= 6x \\ x^2 - 6x + 5 &= 0 \\ (x - 1)(x - 5) &= 0 \\ x &= 1, 5 \end{aligned}$$


Test

$x = -1$	$x = \frac{1}{2}$	$x = 2$	$x = 6$
$\frac{(-1)^2 + 5}{-1} > 6$	$\frac{(\frac{1}{2})^2 + 5}{\frac{1}{2}} > 6$	$\frac{4 + 5}{2} > 6$	$\frac{36 + 5}{6} > 6$
$\times$	$\checkmark$	$\times$	$\checkmark$

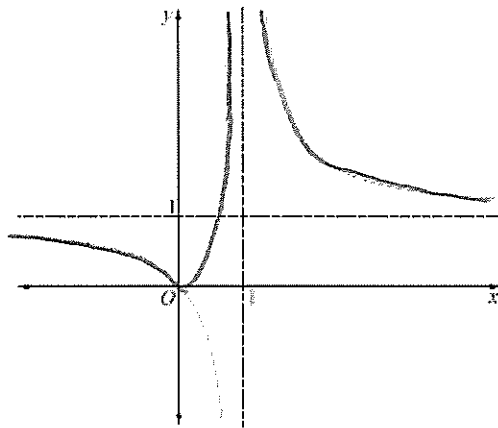
$$\therefore 0 < x < 1, x > 5$$

- c. The diagram shows the graph of the function $f(x) = \frac{x}{x-1}$.

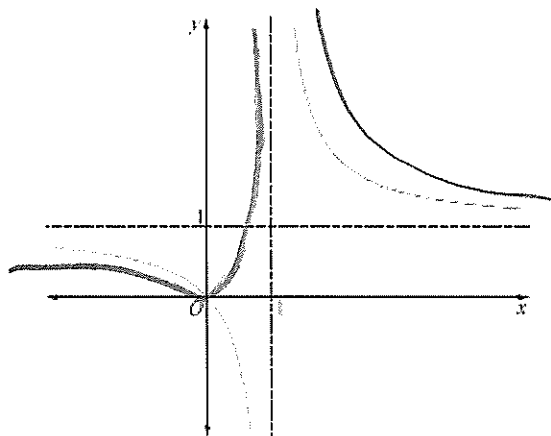


Use the graph of $y = f(x)$ to draw graphs the following transformations on the number planes provided. Clearly show all intercepts and asymptotes.

- (i) $y = |f(x)|$ [3]

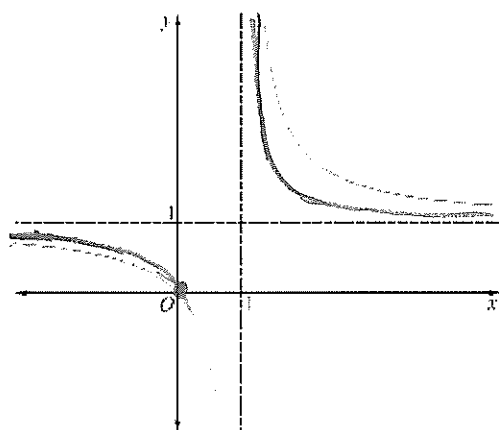


- (ii) $y = [f(x)]^2$ [3]



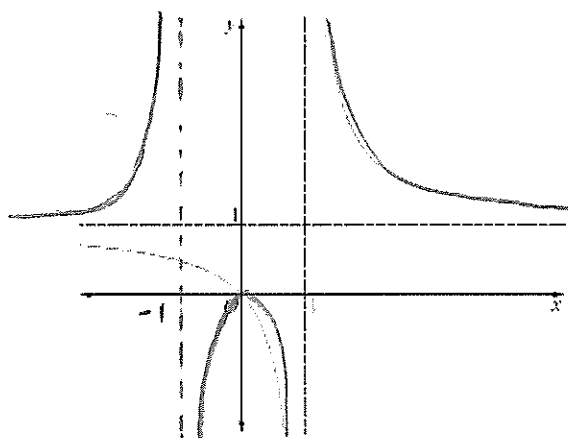
(iii) $y = \sqrt{f(x)}$

[3]



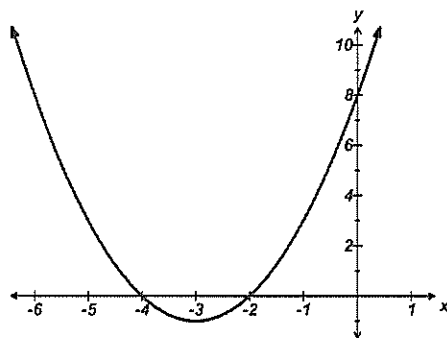
(iv) $y = f(|x|)$

[3]



Question 7 (13 marks)

- a. The graph of $f(x) = x^2 + 6x + 8$ is shown below.



- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. [1]

Horizontal line cuts this graph in more than 1 place
(or) The function is not a one-to-one function.

- (ii) Consider $f(x) = x^2 + 6x + 8$ for $x \geq -3$. Find the equation of $f^{-1}(x)$. [3]

$$f(x) : y = x^2 + 6x + 8$$

$$f^{-1}(x) : x = y^2 + 6y + 8$$

$$x = (y+3)^2 - 1$$

$$x+1 = (y+3)^2$$

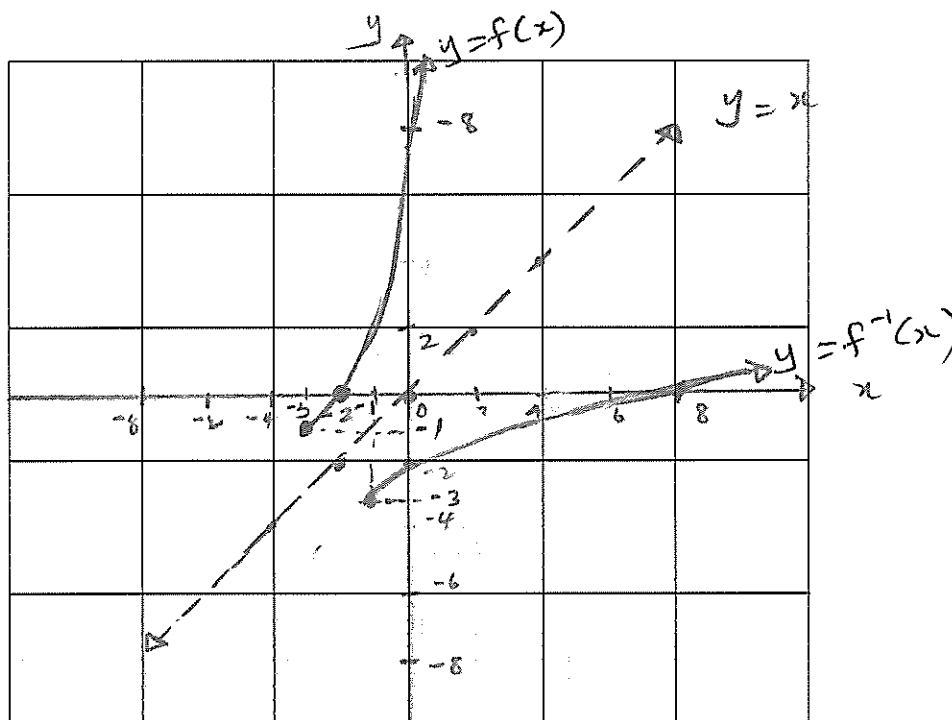
$$y+3 = \pm \sqrt{x+1}$$

$$\therefore y = -3 + \sqrt{x+1}, \text{ since } R_{f^{-1}(x)} : y \geq -3$$

- (iii) State the domain of $f^{-1}(x)$ [1]

$$D_{f^{-1}(x)} = R_{f(x)} = x \geq -1$$

- (iv) Sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$ for $x \geq -3$ on the grid below, clearly showing the important features. [3]



- b. The parametric equation of a graph are:

$$x = 1 - \frac{1}{t}$$

$$y = 1 + \frac{1}{t}$$

- (i) Find the Cartesian equation of the graph. [3]

$$x = 1 - \frac{1}{t} \quad \text{--- (1)} \quad ; \quad y = 1 + \frac{1}{t} \quad \text{--- (2)}$$

From (1) $tx = t - 1$ Substitute $t = \frac{1}{1-x}$ into (2)

$$t - tx = 1$$

$$t(1-x) = 1$$

$$t = \frac{1}{1-x}$$

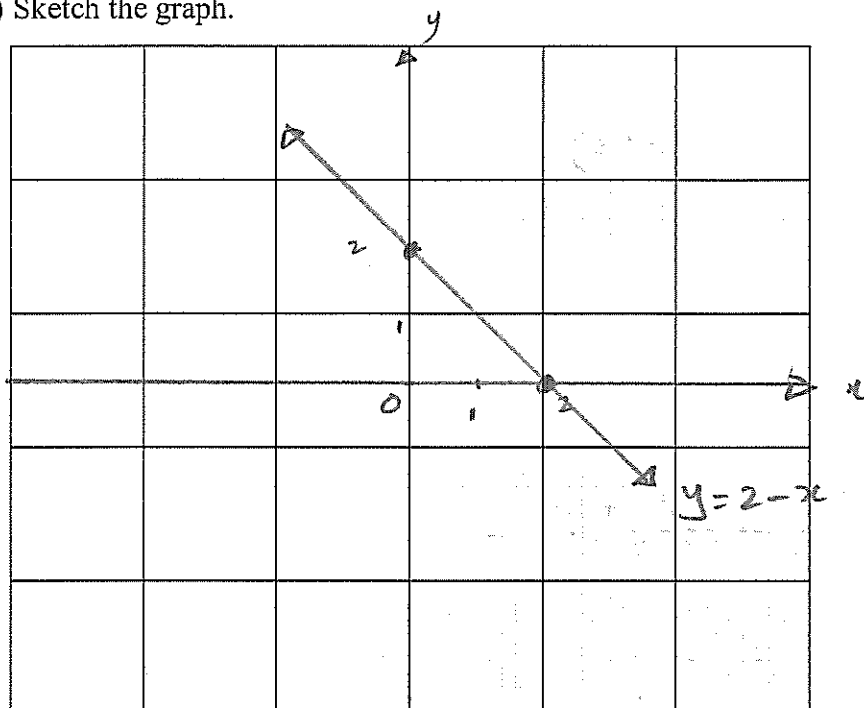
$$y = 1 + \frac{1}{\frac{1}{1-x}}$$

$$y = 1 + 1 - x$$

$$y = 2 - x$$

(ii) Sketch the graph.

[2]



Question 8 (26 marks)

a. If α, β, γ are the roots of $2x^3 - 3x^2 + 6x + 8 = 0$, find the value of:

(i) $\alpha + \beta + \gamma$ [1]

$$= -\frac{b}{a} = \frac{3}{2}$$

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ [1]

$$= \frac{c}{a} = \frac{6}{2} = 3$$

(iii) $\alpha\beta\gamma$ [1]

$$= -\frac{d}{a} = -\frac{8}{2} = -4$$

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]

$$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{3}{-4} = -\frac{3}{4}$$

$$\begin{aligned}
 \text{(v)} \quad & \alpha^2 + \beta^2 + \gamma^2 \quad [3] \\
 & = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\
 & = \left(\frac{3}{2}\right)^2 - 2(3) \\
 & = \frac{9}{4} - 6 = -\frac{15}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi)} \quad & \frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta} \quad [2] \\
 & = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} \\
 & = \frac{-15/4}{-4} = \frac{15}{16}
 \end{aligned}$$

- c. $x^3 + 3x^2 + \alpha x - 1 = 0$ has one root equal to the reciprocal of another.
Find the roots. [3]

$$\text{Roots : } \alpha, \beta, \frac{1}{\beta}$$

$$\text{product of roots : } \alpha = 1$$

$$\therefore \text{ roots} = 1, \beta, \frac{1}{\beta}$$

$$\text{Sum of roots : } 1 + \beta + \frac{1}{\beta} = -3$$

$$\beta + \beta^2 + 1 = -3\beta$$

$$\beta^2 + 4\beta + 1 = 0$$

$$\beta = \frac{-4 \pm \sqrt{12}}{2} = \frac{-4 \pm 2\sqrt{3}}{2}$$

$$= -2 \pm \sqrt{3}$$

$$\therefore \text{ Roots : } 1, \sqrt{3}-2, \frac{1}{\sqrt{3}-2}$$

$$\text{or } 1, -\sqrt{3}-2, \frac{1}{-\sqrt{3}-2}$$

- b. Consider the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$

- (i) Show that $x = 1$ is a zero of $P(x)$. [1]

$$P(1) = 1 - 2 - 5 + 6 = 0$$

$$\therefore x = 1 \text{ is a zero}$$

(ii) Find the other zeros.

[3]

$$\begin{array}{r}
 x^2 - x - 6 \\
 x-1 \overline{) x^3 - 2x^2 - 5x + 6} \\
 \underline{-(x^3 - x^2)} \\
 -x^2 - 5x \\
 \underline{-(-x^2 + x)} \\
 -6x + 6 \\
 \underline{-(-6x + 6)} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x-1)(x^2 - x - 6) \\
 &= (x-1)(x-3)(x+2)
 \end{aligned}$$

Other zeros : $-2, 3$

c. Factorise $2x^3 - 5x^2 - 4x + 12 = 0$ if it has a zero of multiplicity 2 at $x = 2$.

[4]

Roots : $2, 2, \alpha$

$(x-2)^2$ is a factor

$$\text{Sum of roots : } 2 + 2 + \alpha = \frac{5}{2}$$

$$\alpha = -\frac{3}{2}$$

$\therefore (2x+3)$ is a factor

$$\therefore 2x^3 - 5x^2 - 4x + 12 = (x-2)^2(2x+3)$$

d. Let $P(x) = (x+1)(x-3)Q(x) + ax + b$ where $Q(x)$ is a polynomial and a and b are real numbers. The polynomial $P(x)$ has a factor of $x-3$. When $P(x)$ is divided by $x+1$, the remainder is 8.

(i) Find the values of a and b .

[3]

$(x-3)$ is a factor $\therefore P(3) = 0$

$$\therefore 3a + b = 0 \quad \text{--- (1)}$$

When $\div$ by $(x+1)$, remainder = 8 $\therefore P(-1) = 8$

$$\therefore -a + b = 8 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad \therefore 4a = -8$$

$$a = -2$$

Substitute $a = -2$ into $\textcircled{1}$

$$-6 + b = 0$$

$$b = 6$$

$$\therefore a = -2, b = 6$$

(ii) Find the remainder $P(x)$ is divided by $(x+1)(x-3)$.

[1]

$$P(x) = (x+1)(x-3)Q(x) + (-2x+6) \text{ (from (i))}$$

$$\therefore \text{Remainder when divided by } (x+1)(x-3)$$

$$= -2x+6$$

Question 9 (28 marks)

a. Simplify

(i) $n! + (n-1)!$ [2]

$$= n(n-1)! + (n-1)!$$

$$= (n-1)!(n+1)$$

(ii) $\frac{n!}{(n-1)!} \cdot \frac{(n-2)!}{(n+1)!}$ [3]

$$= \frac{n(n-1)!(n-2)!}{(n-1)!(n+1)n(n-1)(n-2)!}$$

$$= \frac{1}{(n+1)(n-1)}$$

b. Siya has five different coloured blocks. She stacks two, three, four or five blocks on top of one another to form a vertical tower.

(i) How many different towers can she form that are three blocks high? [1]

$$5 \times 4 \times 3 \text{ (or)} {}^5P_3 = 60$$

$$= 60$$

(ii) How many different towers can she form in total? [2]

$${}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5$$

$$= 20 + 60 + 120 + 120$$

$$= 320$$

c. (i) In how many different ways can the letters of the word ESTEEM be arranged? [1]

$$\frac{6!}{3!} = 120$$

6 letters, 3 Es

- (ii) In how many ways can the letters of the word ESTEEM be arranged so that the E's are together? [2]

$\boxed{EEE} _ _ _$

$$4! = 24$$

- d. Find the number of ways in which 3 boys and 4 girls line up in a queue so that girls occupy the middle position and exactly one of the two end positions. [3]

$G \dots G \dots B \sim B \dots G \dots G$

$$2 \times {}^4C_2 \times {}^3C_1 \times 4! \times 2! = 1728$$

- e. (i) In how many ways can a committee of 7 be selected from a group of 18 people? [1]

$${}^{18}C_7 = 31824$$

- (ii) In how many ways can a committee of 3 men and 4 women be selected if there are 8 men and 10 women in this group? [2]

$${}^8C_3 \times {}^{10}C_4 = 11760$$

- (iii) In how many ways can a committee of 3 men and 4 women be selected from this group if particular man and a particular woman can't be in the committee together? [3]

Total ways — ways both chosen

$$11760 - {}^7C_2 \times {}^9C_3 = 9996$$

- e. Six men and six women are to be seated at a round table.

In how many ways can they be seated:

- (i) if there are no restrictions? [1]

$$11! = 39916800$$

- (ii) if men and women alternate? [2]

$$5! \times 6!$$

$$= 86400$$

(iii) if two particular people are not to sit together? [3]

11 groups

$$2 \text{ sit together} = 10! \times 2! = 7257600$$

$$\therefore \text{Not sit together} = 11! - 10! \times 2!$$

$$= 10! (11 - 2) = 9 \times 10! = 32659200$$

(iv) if three particular people are to sit together? [2]

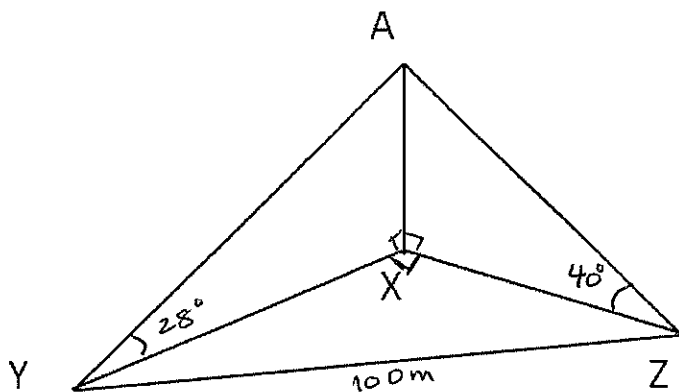
10 groups

$$9! \times 3!$$

$$= 2177280$$

Question 10 (15 marks)

a.



The tower AX, of height h metres, is built on flat ground. X, Y and Z are points on the same horizontal plane.

From Y and Z, the angles of elevation to the top of the tower are 28° and 40° respectively. Give that $YZ = 100\text{m}$ and $\angle YXZ = 90^\circ$,

(i) show that $XY = h \cot 28^\circ$ and find a similar statement for XZ. [2]

In $\triangle AX Y$

$$\tan 28^\circ = \frac{h}{XY}$$

$$XY = \frac{h}{\tan 28^\circ}$$

$$XY = h \cot 28^\circ$$

Similarly, using $\triangle AX Z$

$$XZ = h \cot 40^\circ$$

(ii) find the height of the tower to the nearest metre.

[3]

Using Pythagoras' theorem in $\triangle XYZ$

$$YZ^2 = XY^2 + XZ^2$$

$$(100)^2 = (h \cot 28^\circ)^2 + (h \cot 40^\circ)^2$$

$$= h^2 \cot^2 28^\circ + h^2 \cot^2 40^\circ$$

$$(100)^2 = h^2 (\cot^2 28^\circ + \cot^2 40^\circ)$$

$$h = \frac{100}{\sqrt{\cot^2 28^\circ + \cot^2 40^\circ}}$$

$$= 44.913 \dots$$

height of tower = 45 metres.

b. Find the inverse of $y = 3 \log_e(2x + 5) - 4 = f(x)$

[2]

$$f^{-1}(x) : x = 3 \log_e(2y + 5) - 4$$

$$\frac{x+4}{3} = \log_e(2y+5)$$

$$e^{\frac{x+4}{3}} = 2y+5 \Rightarrow y = \frac{1}{2} \left(e^{\frac{x+4}{3}} - 5 \right)$$

c. Solve $x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 = 0$ if it has a root of multiplicity 4.

[4]

$$P(x) = x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 \quad (\text{root of multiplicity 4})$$

$$P'(x) = 5x^4 + 8x^3 - 6x^2 - 16x - 7 \quad (\text{same root of multiplicity 3})$$

$$P''(x) = 20x^3 + 24x^2 - 12x - 16 \quad (\text{--- --- } x^2)$$

$$P'''(x) = 60x^2 + 48x - 12 \quad (\text{--- --- --- } x^1)$$

$$\text{Roots of } 60x^2 + 48x - 12 = 0$$

$$5x^2 + 4x - 1 = 0$$

$$(5x-1)(x+1) = 0$$

$$x = \frac{1}{5}, -1$$

$$P(-1) = (-1)^5 + 2(-1)^4 - 2(-1)^3 - 8(-1)^2 - 7(-1) - 2$$

$$= 0$$

$\therefore x = -1$ is the zero of multiplicity 4

$\therefore (x+1)^4$ is a factor

$$\therefore P(x) = (x+1)^4(x-2) \quad \text{By observation}$$

$$\therefore x = -1, -1, -1, -1, 2$$

- d. A hostel has four vacant rooms. Each room can accommodate a maximum of four people. In how many ways can six people be accommodated in the four rooms? [3]

$$\text{Total number of ways (no maximum)} = 4^6 = 4096$$

$$\text{Number of ways with max 4} = \text{Total} - (\text{ways to have 5} + \text{ways to have 6})$$

$$\text{No. of ways to have 5 in one room} = {}^6C_5 \times {}^4C_1 \times {}^3C_1 = 72$$

$$\text{No. of ways to have 6 in one room} = 4$$

$$\therefore \text{No. of ways with 4 maximum in a room.}$$

$$= 4096 - (4 + 72)$$

$$= \underline{4020}$$

End of Examination